

Static vs Contextual Embeddings for Lexical Ambiguity: A Word-in-Context Evaluation Method and Results Appendix

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Project repository: <https://github.com/Mr-Dark-debug/static-contextual-ambiguity>

1 Research question and scope

This project asks whether a context-sensitive target representation distinguishes same-sense from different-sense uses more effectively than static alternatives. It compares type-level GloVe, a lexical GloVe context mean, and an unfine-tuned BERT-base target representation. It is a controlled representation probe, not a claim about fine-tuned task state of the art.

2 Dataset provenance and audit

Our examples come from the WiC files in SuperGLUE v2. The archive is available at <https://dl.fbaipublicfiles.com/glue/superglue/data/v2/WiC.zip>. The WiC license is CC BY-NC 4.0.

Table 1: Validated release counts and span diagnostics.

Split	Rows	Positive	Negative	Unlabeled	Lemma mismatch	Repeated surface
Train	5,428	2,714	2,714	0	1,852	93
Validation	638	319	319	0	251	1
Test	1,400	0	0	1,400	521	20

We checked the number of rows, the target positions, and the IDs in each split. Every marked target covers at least one character, and no ID is repeated within its split. Four training pairs appear again with the sentences reversed. We found none of these repeated pairs in validation or across the two splits. We also checked the test rows, but their labels are unavailable, so we did not calculate a test score.

3 Representations

3.1 Static baselines

We used the uncased GloVe 6B vectors with 300 dimensions for both static conditions. The vectors are from the <https://nlp.stanford.edu/projects/glove/> release. In the target-only condition, both sentences use the vector for the same WiC lemma. Its cosine similarity is therefore 1 whenever GloVe contains that lemma. We found 18 training lemmas without vectors and filled those missing scores with the median valid score from training.

For the context condition, we took word tokens on either side of the marked target, left the target out, and averaged the available GloVe vectors. We tested windows of two, five, and ten words on each side, plus the whole sentence. When no context vector was available on one side, we used the lemma vector for that side instead. This happened once per window in training and did not happen in validation.

3.2 Contextual target

The contextual condition uses the pretrained `google-bert/bert-base-uncased` checkpoint, available at <https://huggingface.co/google-bert/bert-base-uncased>. We did not update its weights. The tokenizer reports character positions for each word piece. We selected the pieces that overlap the marked target, averaged their vectors, and saved the target representation from every BERT layer. The mean of the last four layers was one of the configurations compared on training data.

4 Selection and evaluation protocol

We kept model selection inside the training split. Seed 2026 produced 4,342 tuning pairs and 1,086 internal holdout pairs, with balanced labels in both. Each candidate received a similarity threshold fitted on the tuning pairs for macro F1. We compared candidates on the holdout pairs by

macro F1, then by accuracy and ROC-AUC if needed, with a preset simpler choice for exact ties. After choosing the GloVe context window and BERT representation, we fitted their final thresholds using all 5,428 training pairs. We had already checked the validation text for data quality, but its labels were used for scoring only after those choices were fixed.

Table 2: Training-only candidate selection.

Family	Candidate	Tune macro F1	Holdout macro F1	Selected
Static context	glove_context_2	0.5840	0.5961	yes
Static context	glove_context_5	0.5883	0.5928	no
Static context	glove_context_10	0.5917	0.5863	no
Static context	glove_context_sentence	0.5930	0.5845	no
BERT	layer_12	0.6789	0.7145	no
BERT	mean_last_four	0.6891	0.7247	yes

5 Primary validation results

A similarity at or above the threshold counts as a same-meaning prediction. The same-meaning label is positive for precision and recall. We estimated the intervals by resampling the 638 validation pairs 1,000 times with a fixed seed.

Table 3: Primary results on 638 balanced validation pairs.

System	Threshold	Accuracy	Accuracy 95% CI	Macro F1	Precision	Recall	ROC-AUC
GloVe target	1.0000	0.5000	[0.4639, 0.5392]	0.3333	0.5000	1.0000	0.5000
GloVe context (± 2)	0.6506	0.5549	[0.5156, 0.5956]	0.5511	0.5670	0.4639	0.5800
BERT mean last four	0.5813	0.6708	[0.6364, 0.7085]	0.6673	0.6416	0.7743	0.7163

The selected BERT condition answered 428 pairs correctly. GloVe with nearby words answered 354 correctly. Both got 250 pairs right. BERT alone got 178 right, GloVe alone got 104 right, and neither got the remaining 106 right.

Table 4: Paired primary-system differences with 95% intervals.

Contrast	Metric	Estimate	95% interval
Target GloVe minus nearby GloVe	Accuracy	-0.0549	[-0.1176, 0.0094]
Target GloVe minus nearby GloVe	Macro F1	-0.2178	[-0.2661, -0.1710]
Target GloVe minus nearby GloVe	ROC-AUC	-0.0800	[-0.1273, -0.0378]
Target GloVe minus BERT	Accuracy	-0.1708	[-0.2195, -0.1238]
Target GloVe minus BERT	Macro F1	-0.3340	[-0.3739, -0.2953]
Target GloVe minus BERT	ROC-AUC	-0.2163	[-0.2588, -0.1775]
Nearby GloVe minus BERT	Accuracy	-0.1160	[-0.1693, -0.0658]
Nearby GloVe minus BERT	Macro F1	-0.1161	[-0.1690, -0.0664]
Nearby GloVe minus BERT	ROC-AUC	-0.1363	[-0.1912, -0.0811]

For accuracy, macro F1, and ROC-AUC, the paired intervals for GloVe context minus BERT lie below zero. That is evidence of a BERT advantage for these validation examples and these settings. It does not establish the same result for every dataset.

6 Secondary BERT layer analysis

We checked each BERT layer as a secondary analysis. Every layer received its own threshold from training data. The validation scores below describe those layers but were not used to change our primary BERT choice.

Table 5: Individual BERT target layers.

Layer	Threshold	Accuracy	Macro F1	ROC-AUC
1	0.8546	0.5705	0.5685	0.5900
2	0.8145	0.6019	0.6014	0.6187
3	0.7590	0.6113	0.6113	0.6357
4	0.7330	0.6317	0.6314	0.6575
5	0.6949	0.6458	0.6451	0.6768
6	0.6564	0.6442	0.6419	0.6853
7	0.6299	0.6489	0.6467	0.6946
8	0.6039	0.6599	0.6558	0.7036
9	0.5728	0.6661	0.6624	0.7093
10	0.5625	0.6599	0.6543	0.7197
11	0.5947	0.6536	0.6508	0.7096
12	0.4942	0.6364	0.6308	0.6971

The individual-layer accuracy rises overall through layer 9 and then drops by layer 12. Layer 10 has the highest ROC-AUC among single layers. The pattern is consistent with using information from later layers, although our last-four choice had already been made on training data.

7 Error analysis

The 106 pairs missed by both main systems contain 54 that the GloVe target diagnostic also missed and 52 that it happened to answer correctly. BERT made 210 mistakes in all. Of these, 33 were very close to its threshold, within 0.025, while 53 were among the quarter with the largest distances from it.

BERT recalled 77.43% of same-meaning pairs and 56.74% of different-meaning pairs. The corresponding figures for GloVe context were 46.39% and 64.58%. Among our saved examples, BERT separated an anatomical *ball/balls* pair that has the same meaning. It placed different meanings of *engrave* and *map* close together. It also brought together the same meaning in *rust remover* and *paint remover*. These examples help us inspect the mistakes, but they cannot establish a cause for them.

8 Reproducibility record

The full BERT pass ran on an NVIDIA GeForce RTX 3050 Ti Laptop GPU. The repository includes our locked dependencies, configuration, commands, and saved result files for checking the analysis.

9 Limitations

This experiment uses one English benchmark and one frozen BERT base model. The bootstrap intervals vary the sampled validation pairs; they do not cover every choice we made while designing the study. The GloVe average leaves out word order and syntax, and one cutoff may hide useful differences between words. The unpublished test labels prevented us from calculating a test result.

10 Author contributions and use of AI tools

We selected WiC, set up the three conditions, chose the pretrained GloVe and BERT models, and made the configuration and threshold decisions from training data. We ran the dataset checks and the evaluation, and we checked the reported numbers and graphs against our saved results. We did not train or fine-tune model weights.

We wrote the research content, poster explanations, and interpretations together. We used Canva while planning the poster layout and GLM to help with some LaTeX layout code. OpenAI Codex assisted us as a coding partner, including debugging, generating the LaTeX tables from our result files, and checking the PDFs. It also suggested more formal phrasing in parts of this appendix. We read those suggestions, corrected the wording, and checked that the tables and claims still match our own experiment. The explanatory text in Sections 1–9 contains this AI-assisted phrasing. We remain responsible for the final code, text, figures, and submission.

References

- [1] J. Camacho-Collados and M. T. Pilehvar. From Word to Sense Embeddings: A Survey on Vector Representations of Meaning. *JAIR*, 2018. <https://doi.org/10.1613/jair.1.11259>.
- [2] M. T. Pilehvar and J. Camacho-Collados. WiC: The Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations. *NAACL*, 2019. <https://doi.org/10.18653/v1/N19-1128>.

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